

AIS5103 homework 3 (due Friday 23 Oct 2026)

1. Read Chapter 10 of Bishop's book, "Deep Learning," pp. 287-301, up to the VGG architecture. Summarize your reading in no more than one A4-size page. Handwritten notes are preferred.
2. In this programming problem, we implement a simple physics-informed neural network (PINN). We consider predicting a function $x(t)$ in the time interval $0 \leq t \leq t_{\max}$. We know that $x(t)$ describes the position of a damped oscillator measured in a physics experiment. It is modeled by the second-order ordinary differential equation, the Newton equation of motion,

$$m \frac{d^2 x}{dt^2} + \gamma \frac{dx}{dt} + kx = 0$$

where m is the mass of the particle, γ is the damping parameter, and k is the spring constant. For computational simplicity, we assume $m = 1, \gamma = 0.1, k = 1$, in some units. A student performed the experiment and recorded the position x as a function of time t . Let's say, out of laziness, we only know the values at two points: at $t = 0$ we have $x(0) = 1.0$, and at $t = 1.0$ we find $x(1.0) = -3.0$. (a) Set up a PINN to find the solution $x(t)$ for t in the interval $[0, 15.0]$. Clearly state the rationale for your hyperparameter choices. (b) Compare the neural network solution with the exact solution. (c) Comment on how the PINN formulation would change if the student reports more experimental data points, or if some of the model parameters, such as the damping rate γ , are not known.

3. Bishop, exercise 10.9, p. 323. Consider a convolution of the form (10.2)

$$C(j, k) = \sum_l \sum_m I(j + l, k + m) K(l, m)$$

here C is the feature map, I is image pixel intensities, and K is a learnable kernel. Suppose that the kernel is separable so that

$$K(l, m) = F(l)G(m)$$

for some functions $F(\cdot)$ and $G(\cdot)$. (a) Show that instead of performing a single two-dimensional convolution, it is now possible to compute the same answer using two one-dimensional convolutions, thereby significantly improving computational efficiency. (b) For an $M \times M$ kernel and an $N \times N$ image, state the order of computational complexities for the two cases.

4. Backpropagation through time for a simple RNN. Consider the following scalar recurrent neural network:

$$h_t = \tanh(W h_{t-1} + U x_t), \quad y_t = V h_t,$$

where x_t is the input, h_t is the hidden state, and W , U , and V are scalar parameters (1 by 1 matrices). The initial hidden state h_0 is given. Note that all of these quantities are scalar real numbers.

Unfold the RNN for two time steps, $t = 1, 2$. The loss is evaluated at the final time step:

$$E = \frac{1}{2} (y_2 - \hat{y})^2,$$

where $\hat{y}$ is the target value.

Using backpropagation through time, derive an explicit expression for $\partial E / \partial W$. Your final answer should explicitly show the two contributions to the gradient arising from the occurrences of W at $t = 1$ and $t = 2$.

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